THE HENSELIZATION FUNCTOR ON LOCAL RINGS AGREES WITH THE HENSELIZATION OF THE ASSOCIATED PAIR

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Allowed standard prerequisites (editorial).

- Construction of the henselization of a pair. Lemma \ref{lemma-henselization}, cited at proof lines 3118 and 3129 and 3132-3134. Provides the ring whose maximal ideal and colimit description are studied.
- The maximal ideal extends to a maximal ideal contained in the Jacobson radical, and the residue field is unchanged. Lemma \ref{lemma-henselization-flat}, cited at proof lines 3119-3120 and 3136. Shows that the henselization of the pair is a local ring with the expected maximal ideal and residue field.
- The construction of the henselization of a local ring as a colimit of etale algebras. Algebra, Lemma \ref{algebra-lemma-henselization}, cited at proof lines 3127-3128. Allows the two constructions to be compared directly.
- Uniqueness of the henselian local ring which is a filtered colimit of etale algebras with prescribed residue field. Algebra, Lemma \ref{algebra-lemma-uniqueness-henselian}, cited at proof lines 3137-3138. Gives the canonical isomorphism in the second proof.

Source reference. Original lemma 0A03: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12621–12621. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0A02: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12620–12620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AGU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13207–13207. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5839–5839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 08HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10791–10791. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The functor of Lemma 0A02 associates to a local ring* $(A, \mathfrak{m})$ *its henselization.*

Proof. Let $(A^h, \mathfrak{m}^h)$ be the henselization of the pair $(A, \mathfrak{m})$ constructed in Lemma 0A02. Then $\mathfrak{m}^h = \mathfrak{m}A^h$ is a maximal ideal by Lemma 0AGU and since it is contained in the Jacobson radical, we conclude A^h is local with maximal ideal $\mathfrak{m}^h$. Having said this there are two ways to finish the proof.

First proof: observe that the construction in the proof of Algebra, Lemma 04GN as a colimit is the same as the colimit used to construct A^h in Lemma 0A02. Second proof: Both the henselization $A \to S$ and $A \to A^h$ of Lemma 0A02 are local ring homomorphisms, both S and A^h are filtered colimits of étale A-algebras, both S and A^h are henselian local rings, and both S and A^h have residue fields equal to $\kappa(\mathfrak{m})$ (by Lemma 0AGU for the second case). Hence they are canonically isomorphic by Algebra, Lemma 08HT. □